THREE-BLOCK FIRST-ORDER DEFINITIONS FOR PSEUDOVARIETIES OF FINITE SEMILATTICE-BASED ALGEBRAS

LUCY HAM, MARCEL JACKSON

Editorial note. Full source-specific supporting argument is retained for this target; other statements appearing as support do not create additional counted tasks.

Editorial note. Public 134512 is thm:FO, a later particular non-finitely-based algebra. This target uniformly constructs the quantifier-bounded definition for any finite family of semilattice-based finite algebras.

Allowed standard prerequisites (editorial).

- Standard subject prerequisites. Standard graph/probability and finite or stable model-theory definitions and classical named results cited by the authors. All source-specific local proofs are retained in the original order; named classical inputs are not recursively re-proved.

Source reference. Original section 1.3: sources/ham-jackson-pseudovarieties-2212.02653v4/source0.tex, source lines 252–252. Original cross-reference anchor restored for the already selected source excerpt. This is a layout/reference repair only; the original proof selection and mathematical scope are unchanged.

Source reference. Original section 7: sources/ham-jackson-pseudovarieties-2212.02653v4/source0.tex, source lines 988–988. Original cross-reference anchor restored for the already selected source excerpt. This is a layout/reference repair only; the original proof selection and mathematical scope are unchanged.

Source reference. Original lemma 6.4: sources/ham-jackson-pseudovarieties-2212.02653v4/source0.tex, source lines 934–934. Original cross-reference anchor restored for the already selected source excerpt. This is a layout/reference repair only; the original proof selection and mathematical scope are unchanged.

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1. Preliminaries

1.1. **Varieties and other algebraic concepts.** We direct the reader to a typical universal algebra text such as [5] or [12] for deeper explanation around the following concepts. A reader familiar with the area can skip the section entirely.

The algebraic focus of the article is primarily concerned with theory of varieties and to a lesser extent quasivarieties and universal Horn classes, and their restriction to finite algebras. Recall from the introduction that a variety of algebras is any class $\mathcal{V}$ of algebras of the same signature that is defined by some set (possibly infinite) of equations, or equivalently by Birkhoff's preservation theorem if it is closed under the class operators H, S and P. We now give some basic examples of classes encountered in the paper.

Example 1.1. *The variety of semilattices is the class of algebras with a single binary operation $\wedge$ defined by idempotence, commutativity and associativity*:

$$x \wedge x \approx x, \quad x \wedge y \approx y \wedge x, \quad x \wedge (y \wedge z) \approx (x \wedge y) \wedge z,$$

each universally quantified. Equivalently, the variety of semilattices is equal to $\mathsf{HSP}(\mathbf{S}_\wedge)$*, where* $\mathbf{S}_\wedge$ *denotes the 2-element semilattice (which can be represented by interpreting $\wedge$ as the usual number-theoretic multiplication on the two-element set $\{0, 1\}$).*

We will say that an algebra is *semilattice-based* if its fundamental operations include a binary operation satisfying the semilattice axioms.

When $\mathbf{A}$ is finite, then the finite members of $\mathsf{HSP}(\mathbf{A})$ can be obtained by taking only finitary direct products P_{fin}; in other words we need only consider the pseudovariety $\mathsf{HSP}_{\text{fin}}(\mathbf{A})$. While the equational theory of $\mathbf{A}$ continues to define the finitely generated pseudovariety $\mathsf{HSP}_{\text{fin}}(\mathbf{A})$ amongst finite algebras (but possibly requiring infinitely many equations), pseudovarieties cannot in general be defined by equations. Instead, pseudoequations are the most common approach: equalities between limits of sequences of terms rather than between terms (see Almeida [2] for example).

Example 1.2. *The pseudovariety of $\mathcal{J}$-trivial monoids is the class of finite monoids satisfying*

$$\forall a \forall b \forall x_1 \forall y_1 \forall x_2 \forall y_2 \; x_1 a y_1 \approx b \;\; \& \;\; x_2 b y_2 \approx a \to a \approx b.$$

and can also be axiomatised by pseudoequations $x^\omega \approx x^{\omega+1}$ and $(xy)^\omega \approx (yx)^\omega$, where x^ω denotes the implicit operation defined as the limit $\lim_{n\to\infty} x^{n!}$, the limit of an eventually constant sequence in any finite monoid.

We will write $\text{Mod}_{\text{fin}}(\Sigma)$ to denote the *finite* models of a sentence, or set of sentences Σ in some logical framework, and say that a class $\mathcal{K}$ of finite algebras is *first order definable* (amongst finite models) if there is a first order sentence Σ such that $\mathcal{K} = \text{Mod}_{\text{fin}}(\Sigma)$. From Example 1.2, the $\mathcal{J}$-trivial monoids are first order definable, as $\text{Mod}_{\text{fin}}(\Sigma)$, where Σ is the finite set consisting of the usual equational axioms for monoids along with the implication displayed in the example.

As described in Subsection 1.3, a *universal Horn class* is a class of structures (algebras, relational structures, or general structures with both operations and relations), defined by universally quantified conjunctions of Horn clauses. These may also equivalently be described as classes closed under taking isomorphic copies induced substructures (S again), direct products over nonempty sets of models P^+, and ultraproducts P_u.

Example 1.3. *Simple graphs may be considered as symmetric loopless digraphs, and the class of all such structures is a universal Horn class defined by*

$$\neg(x, x) \in R, \qquad \neg(x, y) \in R \vee (y, x) \in R.$$

This universal Horn class has no finite generator.

As usual, and as noted in Subsection 1.3, when a Horn clause has exactly one positive disjunct, then we can rewrite it as an implication: so $\neg(x,y) \in R \vee (y,x) \in R$ becomes $(x,y) \in R \to (y,x) \in R$; the implication in Example 1.2 is another example. Sentences of this form are usually called quasiequations, and classes definable by quasiequations are known as quasivarieties. The distinction between quasivarieties and the more general universal Horn classes is minimal, as they differ only in that for quasivarieties, the direct product closure condition is extended to include the direct product over the empty set of structures. This degenerate case of direct product returns the one element total structure (any relations in the signature are total on this one point). This one-element structure is the only difference between a quasivariety and a universal Horn class, so from a computational and axiomatisability perspective they are not significantly different.

Recall that a class is *locally finite* if for every n, the n-generated substructures are finite. For classes of finite objects this notion is degenerate, but the next notion is not. A class is *uniformly local finite* if for all n, there exists m such that the n-generated substructures have size at most m. In most cases, what we really use is the less well-known concept *regularly locally finite*, which means that there are only finitely many n-generated substructures up to isomorphism, all finite. In finite signatures however, uniform and regular local finiteness are equivalent, see Bezhanishvili [9] (noting that we have bypassed consideration of whether the class in question is hereditary by including "substructures" in the definition). All classes of relational structures are uniformly locally finite, as there is no generation power. The following lemma is folklore and we omit the easy proof.

Lemma 1.4. *Let $\mathcal{K}$ be a uniformly locally finite class of algebras of the same finite signature. Then the pseudovariety generated by $\mathcal{K}$ is also uniformly locally finite.*

The next lemma is usually called Birkhoff's Finite Basis Theorem after Garrett Birkhoff [10]; versions of it can be found in any universal algebra text; see Theorem 5.27 of [5], Theorem 4.2 of Burris and Sankappanavar [12], or Theorem 7.16 of Freese, McKenzie, McNulty and Taylor [20] for example. We give a slight enhancement to the usual presentation, and for this reason only we sketch the proof, even if it is not more than some observations made along the way of the standard proof. The key addition is that the replacement rule alone is sufficient in syntactic equational inference; see Section 7.2 of [20] for a thorough treatment of syntactic equational inference.

Birkhoff's Finite Basis Theorem 1.5. *Let $\mathcal{K}$ be a uniformly locally finite class of algebras in a finite signature, let k be a positive integer, and let $\vec{x}$ abbreviate a k-tuple of variables $x_0, \ldots, x_{k-1}$. Then either $\mathcal{K}$ is trivial and satisfies $x \approx y$ or there is a finite set of terms $T_{\mathrm{BFB}} := \{u_i(\vec{x}) \mid i = 1, \ldots, \ell\}$ with the following properties*:

(1) T_{BFB} is closed under taking subterms;
(2) for every fundamental operation f of arity n, and every $i_1, \ldots, i_n \leq \ell$ there is precisely one $i' \leq \ell$ for which

$$\mathcal{K} \models f(u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x})) \approx u_{i'}(\vec{x});$$

(let the set of these satisfied equations be denoted Σ_{BFB};)
(3) for every term $t(\vec{x})$ there is a unique $i \leq \ell$ such that $\mathcal{K} \models t(\vec{x}) \approx u_i(\vec{x})$ and
(4) whenever $\mathcal{K} \models t(\vec{x}) \approx u_i(\vec{x})$ there is a proof of $t(\vec{x}) \approx u_i(\vec{x})$ from Σ_{BFB} that proceeds by a sequence of exact replacements (no variable substitutions are required).

In particular, Σ_{BFB} is a finite basis for the k-variable equational theory of $\mathcal{K}$.

Proof. Because $\mathcal{K}$ is uniformly locally finite, the k-generated free algebra $\mathbf{F}$ for $\mathsf{HSP}(\mathcal{K})$ is finite. The properties described are then just a description of a canonical presentation of $\mathbf{F}$ by its operation tables, with item (4) corresponding to the evaluation of $t(\vec{x})$ in the algebra $\mathbf{F}$ at the free generators. For more detail, inductively construct T_{BFB} as a list of representatives of the elements of $\mathbf{F}$, starting with the variable symbols $x_0, \ldots, x_{k-1}$ as free generators, which are distinct elements of $\mathbf{F}$ and the base of case of the generation of T_{BFB}. Now iterate the following process; for each operation f (of arity n, say), and each selection of n elements from the list $u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x})$ of so-far constructed terms, check to see if $f(u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x}))$ represents a new element of $\mathbf{F}$: if so, add it to the list; otherwise ignore it. We have T_{BFB} at the inevitable termination of this process ($\mathbf{F}$ is finite): when no new elements are added to the list, for all operations and all selections of inputs to those operations from the list so far constructed. Properties (1), (2) and (3) follow very easily from this construction. For item (4) observe that when an arbitrary term $t(x_0, \ldots, x_{k-1})$ is evaluated in $\mathbf{F}$ at the free generators $x_0, \ldots, x_{k-1}$, the evaluation process constitutes a reduction to an element in T_{BFB}. More precisely, we may proceed as follows: take a smallest subterm of $t(x_0, \ldots, x_{k-1})$ that does not appear in T_{BFB}. Because the variables are in T_{BFB}, this subterm must be of the form $f(u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x}))$ for some fundamental operation f and some terms $u_{i_1}(\vec{x})$, ..., $u_{i_n}(\vec{x}))$ in T_{BFB}. By (2) there is a unique $i' \leq \ell$ such that $\mathcal{K} \models f(u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x})) \approx u'(x_0, \ldots, x_{k-1})$ and this law is in Σ_{BFB}. Replace the subterm $f(u_{i_1}(\vec{x}), \ldots, u_{i_n}(\vec{x}))$ by the new term $u'(x_0, \ldots, x_{k-1})$. Repeating this evaluation strategy eventually leads to an element of T_{BFB}. □

Of course, T_{BFB} and Σ_{BFB} depend on both $\mathcal{K}$ and k, and we write $T_{\mathrm{BFB}}(\mathcal{K}, k)$ and $\Sigma_{\mathrm{BFB}}(\mathcal{K}, k)$ when we wish to make this clear.

1.2. **First order reductions.** We now recall the definition of a first-order reduction; see [29] or [47] for a more complete treatment. In the standard definition for computational complexity we are allowed recourse to a strict linear order $<$ and work entirely within relational signatures. Every operation of arity n can be considered as a relation of arity $n+1$ using its graph.

For $n \in \mathbb{N} \cup \{0\}$, purely relational signatures $\mathcal{R}$ and $\mathcal{S}$ and variables $\vec{x} = x_0, \ldots, x_{n-1}$, a first-order $\mathcal{R} \cup \{<\}$-formula $\varphi(\vec{x})$ determines a family of n-ary relations on any $\mathcal{R} \cup \{<\}$-structure $\mathbb{B}$ in the following way. First, the solution set of $\varphi(\vec{x})$ is an n-ary relation. Second, for any fundamental relation $s \in \mathcal{S}$ of arity k, a kn-ary formula $\psi_s(\vec{x}_1, \ldots, \vec{x}_k)$ (where $\vec{x}_i$ denotes $x_{i,0}, \ldots, x_{i,n-1}$) in the language of $\mathcal{R} \cup \{<\}$ determines a k-ary relation on n-tuples: a k-ary relation on B^n. The family of formulæ

$$\{\varphi(\vec{x})\} \cup \{\psi_s(\vec{x}_1, \ldots, \vec{x}_k) \mid s \in \mathcal{S}\}$$

defines an $\mathcal{S}$-structure $\mu(\mathbb{B})$, whose universe U is the solution set of $\varphi(\vec{x})$, as a subset of B^n, and where the relations on $\mu(\mathbb{B})$ are the restriction to U of the solution sets of $\psi_s(\vec{x_1}, \ldots, \vec{x_k})$, for each $s \in \mathcal{S}$. These formulæ form an n-ary *first order query*.

We will also explore a one-to-many variation of this idea. In this formulation we have a first order query, as just defined but allow some additional free variables $y_1, \ldots, y_k$ in all formulæ. Each evaluation of these variables in some structure $\mathbb{A}$, say $(y_1, \ldots, y_k) \mapsto (a_1, \ldots, a_k)$, will give rise to a first order query with parameters $a_1, \ldots, a_k$. The intention is that we have a reduction from problem $\mathcal{P}$ to $\mathcal{Q}$, where structure $\mathbb{A}$ has a property $\mathcal{P}$ if and only if every evaluation $(y_1, \ldots, y_k) \mapsto (a_1, \ldots, a_k)$ in the universe A has property $\mathcal{Q}$. In particular, we observe that if property $\mathcal{Q}$ is first order, then so also is property $\mathcal{P}$, as we are simply asserting that the structure defined by the first order query has first order property $\mathcal{Q}$ for all choices of $a_1, \ldots, a_k$.

2. Toolkit: principal congruences and subdirect irreducibility

In this first toolkit section we consider the precise complexity of several computational problems relating to the subdirect decomposition of algebras. Previous contributions due to Bergman and Slutzki [6, 7, 8] give much of the general picture, though we solve one open problem in their work, and find improved complexity for a range of important classes of interest. The notion of definable completely meet irreducible congruences (introduced in Definition 2.5) and their subsequent properties are central to results in Section 5 and 7.

Recall that an algebra is *subdirectly irreducible* if there is a pair of distinct elements $a \neq b$ that is contained in the principal congruence of all pairs. Equivalently, there is a unique minimum nontrivial congruence: the *monolith*. As every algebra is a subdirect product of its subdirectly irreducible quotients, we obtain a reduction from var-mem$(*, \mathbf{B})$ to the problem var-si-mem$(*, \mathbf{B})$. This reduction can be performed in polynomial time [17], though the authors are unaware of any finer level classification of the complexity of this process. We explore related ideas, presenting one specific case where the subdirectly irreducible quotients may be created as first order queries.

We will be interested in the following computational problems on finite algebras.

Subdirect irreducibility
Instance: a finite algebra $\mathbf{A}$ of finite signature.
Question: is $\mathbf{A}$ subdirectly irreducible?

Principal congruence membership
Instance: a finite algebra $\mathbf{A}$ of finite signature and two unordered pairs $\{a, b\}$ and $\{c, d\}$ taken from A.
Question: is (a, b) in the principal congruence of $\mathbf{A}$ generated by (c, d)?

The principal congruence generated by a pair (c, d) is often denoted $\mathrm{Cg}(c, d)$, so that the question in principal congruence membership can be written "is $(a, b) \in \mathrm{Cg}(c, d)$?"

cong-class
Instance: A pair $(\mathbf{A}, C)$, where $\mathbf{A}$ is an algebra and $C \subseteq A$.
Question: Is there a congruence θ on $\mathbf{A}$ in which C is a congruence class?

Before we state our results we recall the definition of a (directed) graph algebra; see Kelarev [40] for example.

Definition 2.1. Let $\mathbb{G} = \langle G, E \rangle$ be a directed graph; in other words E is a binary relation on the set G. The *graph algebra* of $\mathbb{G}$ is the algebra on the disjoint union $G \cup \{0\}$ with a single binary operation $\cdot$ defined by

$$u \cdot v = \begin{cases} v & \text{if } (u, v) \in E \\ 0 & \text{otherwise.} \end{cases}$$

The following theorem is mostly due to Bergman and Slutzki, though item 3 solves positively the problem stated at [8, p.598], and therefore negatively the problem stated at [8, p.603].

Theorem 2.2. *In any finite algebraic signature containing an operation of arity at least* 2*:*

(1) ([8]) *Principal congruence membership is* NL*-complete.*
(2) ([8]) *Subdirect irreducibility is* NL*-complete.*
(3) *cong-class is* NL*-complete.*

Proof. All three statements are proved by reduction from the undirected graph reachability problem. Items 1 and 2 are [8, Theorem 3.5] (which also considers

the problem of recognising simplicity for a finite algebra; we have no use for this here though). We consider item 3. It is already shown in [8] that cong-class is in NL. We need to show hardness, reducing from the NL-complete problem of st-connectivity, also known as directed graph reachability, and noting that co-NL = NL by the Immerman–Szelepcsényi theorem [29].

Consider a finite directed graph $\mathbb{G} = \langle V; E\rangle$ with distinguished vertices u, v. Let $a, b, 0$ be symbols not in V and construct the graph $\mathbb{G}_{a,b}$ on $V \cup \{a, b\}$ with edges consisting of E along with (a, u), (v, a) and (v, b). We claim that in the graph algebra of $\mathbb{G}_{a,b}$, the subset $\{a, b\}$ is a congruence class if and only if v is not reachable from u. Clearly $\{a, b\}$ is a congruence class if and only if it is the congruence class of the congruence $\mathrm{Cg}(a, b)$, thus it suffices to show that $\{a, b\}$ is a congruence class of $\mathrm{Cg}(a, b)$ if and only if v is not reachable from u.

Assume v is reachable from u. So v is also reachable from a, and then b is reachable from a; let $a = a_0, u = a_1, a_2, \ldots, a_n = v$ be a sequence of vertices leading from a to v. So $(\ldots((aa_1)a_2)\ldots a_n)b = vb = b$ and then $a \equiv b$ implies $b \equiv (\ldots((ba_1)a_2)\ldots a_n)b = 0$. So $\{a, b\}$ is not a congruence class.

Now assume that v is not reachable from u. Let U denote the subset of $V \cup \{a, b\}$ reachable by a nontrivial path from u, and observe that $a, b, v \notin U$. It is now routine to show that $\mathrm{Cg}(a, b)$ consists of the following blocks: $\{a, b\}$, $U \cup \{0\}$ and all singletons from $V \backslash U$.

This proof concerns only the case where there is a single binary operation, but the argument can be transformed trivially into any signature containing at least one operation f of arity ≥ 2 by defining f by $f(a_1, a_2, \ldots) \coloneqq a_1 \cdot a_2$, where $\cdot$ is the operation, and letting all other operations be projections. □

As we shortly demonstrate, a wide range of algebraic settings enjoy a weak congruence condition that improves on Theorem 2.2. Let T_x be the set of all terms in an algebraic signature in variables $\{x, z_1, z_2, \ldots\}$, and in which the variable x explicitly occurs, and let F be a subset of T_x. For an algebra $\mathbf{A}$ and $a, b, c, d \in A$ we write $\{c, d\} \not\hookrightarrow^1_F \{a, b\}$ if $a = b$ or there is a term $t(x, z_1, z_2, \ldots) \in F$ and elements $e_0, e_1, \cdots \in A$ such that

$$\{t^{\mathbf{A}}(c, e_0, e_1, \ldots), t^{\mathbf{A}}(d, e_0, e_1, \ldots)\} = \{a, b\}.$$

Define $\{c, d\} \not\hookrightarrow^n_F \{a, b\}$ if there is a sequence $a = a_0, a_1, a_2, \ldots, a_n = b$ with $\{c, d\} \not\hookrightarrow^1_F \{a_i, a_{i+1}\}$ for each $i = 0, \ldots, n-1$. Finally, write $\{c, d\} \not\hookrightarrow_F \{a, b\}$ if $\{c, d\} \not\hookrightarrow^k_F \{a, b\}$ for some k. (This notation is based on Baker, McNulty and Wang [3], see also the book [20, p. 25].) Maltsev proved that a pair $(a, b) \in \mathrm{Cg}(c, d)$ if and only if $\{a, b\} \not\hookrightarrow_{T_x} \{c, d\}$. A class has *term finite principal congruences* (TFPC) [13] if there a finite set $F \subseteq T_x$ such that $\not\hookrightarrow_{T_x}$ coincides with $\not\hookrightarrow_F$. A class has *definable principal congruences* (DPC) if there exists n and a finite subset $F \subseteq T_x$ with $\not\hookrightarrow_{T_x} = \not\hookrightarrow^n_F$. The DPC property is rather special, but as explained in [13], many familiar classes of algebraic structures have the TFPC property, including semigroups, groups, rings and lattices.

The following generalised version of principal congruence membership is useful before we state and prove the theorem.

Congruence membership
Instance: a finite algebra $\mathbf{A}$ of finite signature, a subset $S \subseteq A^2$ and an unordered pair $\{a, b\}$.
Question: is (a, b) in the congruence of $\mathbf{A}$ generated by S?

Theorem 2.3. *Let $\mathcal{V}$ be a variety of finite signature and with TFPC.*

(0) For any finite algebra $\mathbf{A}$ the congruence membership problem is solvable in L.

As a consequence, each of the following problems are in L:

(1) Principal congruence membership;
(2) Subdirect irreducibility;
(3) cong-class.

Proof of Theorem 2.3. We first explain why the solvability of (1), (2) and (3) in $\mathtt{L}$ follows from the logspace solvability of (0) almost trivially: in each case we can solve in logspace using an oracle for (0), and as $\mathtt{L}^{\mathtt{L}} = \mathtt{L}$ we obtain membership in $\mathtt{L}$. Note that space considerations still only concern the working tape, and not the oracle tape, so that we do not need to concern ourselves with the size of any constructed S written to the oracle, only with the working tape required to perform any calculations needed for that writing; see [11, Section 6.2] for detailed discussion on this.

Item (1) is a specific case of (0), where $S = \{(c, d)\}$.

For item (2), subdirect irreducibility can be decided in $\mathtt{L}$ by systematically enumerating (one-by-one, re-using the same piece of work tape) all pairs $a \neq b$, and in each case systematically enumerating (one-by-one, reusing some new piece of work tape) all $c \neq d$, testing for satisfaction of $(a, b) \in \mathrm{Cg}(c, d)$ using the oracle. Subdirect irreducibility holds if there is a choice of (a, b) such that every pair $c \neq d$ satisfies $(a, b) \in \mathrm{Cg}(c, d)$.

For item (3), observe that a subset $C \subseteq A$ is a congruence class of a congruence if and only if it is a congruence class of the congruence $\mathrm{Cg}(S)$, where $S = C^2 \subseteq A^2$. Because C is trivially a subset of a congruence class of $\mathrm{Cg}(C^2)$, we can decide equality with a congruence class, by selecting an arbitrary $c \in C$, and then systematically verifying $(d, c) \notin \mathrm{Cg}(C^2)$ using the oracle for (0), for each $d \notin C$. Again we have reduced to a special case of (0), where $S = C^2$.

To prove (0), let F be a finite subset of T_x that determines principal congruences in $\mathcal{V}$, let $\mathbf{A}$ be a finite member of $\mathcal{V}$ and $S \subseteq A^2$. For any given pair $(a, b) \in A^2$, if $a = b$ then $(a, b) \in \mathrm{Cg}(S)$ trivially, so we assume that $a \neq b$. Then $(a, b) \in \mathrm{Cg}(S)$ if and only if a is reachable from b in the undirected graph on A whose undirected edges are $\{\{u, v\} \in A^2 \mid S \not\hookrightarrow^1_F \{u, v\}\}$. As F is finite, the relation $\{c, d\} \not\hookrightarrow^1_F \{u, v\}$ is a first order formula for each $c \neq d$ with $(c, d) \in S$. The relation $S \not\hookrightarrow^1_F \{u, v\}$ is simply the disjunction of the formulæ $\{c, d\} \not\hookrightarrow^1_F \{u, v\}$ over all pairs $(c, d) \in S$, and hence is also a first order formula. Thus the edge relation of this graph is itself first order definable from the input subset S and algebra $\mathbf{A}$, and so can be constructed in logspace. Thus (0) holds because undirected graph reachability can be solved in $\mathtt{L}$; see [55, 56]. □

Theorem 2.3 (2) shows that, under the TFPC assumption, the promise of subdirect irreducibility in restricting $\mathsf{var\text{-}mem}(*, \mathbf{B})$ to $\mathsf{var\text{-}si\text{-}mem}(*, \mathbf{B})$ can be verified in logspace. A more useful contribution would be if it gave rise to a logspace reduction from $\mathsf{var\text{-}mem}(*, \mathbf{B})$ to $\mathsf{var\text{-}si\text{-}mem}(*, \mathbf{B})$, however it is not clear to the authors if this is true; even in nondeterministic logspace.

Question 2.4. Is there a reasonably applicable algebraic condition that guarantees the logspace construction of subdirectly irreducible quotients of a finite input algebra $\mathbf{B}$?

It would be interesting if the TFPC condition was sufficient for example or some well known Maltsev condition. We will give two example solutions to Question 2.4: in Proposition 2.8 and in Lemma 6.4. These are strong solutions in the sense that the subdirectly irreducible quotients may be constructed as a first order query (and hence within deterministic logspace), thereby showing that $\mathsf{var\text{-}mem}(*, \mathbf{B})$ and $\mathsf{var\text{-}si\text{-}mem}(*, \mathbf{B})$ are in fact first order equivalent. Neither are "classical" universal algebraic properties however.

Definition 2.5. Let $\mathcal{K}$ be a class of algebras of the same signature. The class $\mathcal{K}$ is said to have *definable completely meet irreducible congruences* (abbreviated to *definable cmi congruences*) if there is a formula $\pi_{x,y}(u,v)$ with four free variables x,y,u,v such that every $\mathbf{A} \in \mathcal{K}$ and every $a,b \in A$ with $a \neq b$, the binary relation $\{(c,d) \in A^2 \mid \mathbf{A} \models \pi_{a,b}(c,d)\}$ defined by the formula $\pi_{a,b}(u,v)$ is a congruence that is maximal with respect to not containing the pair (a,b).

We remind the reader that the property that a congruence θ is maximal with respect to $(a,b) \notin \theta$ for some a,b is equivalent to θ being completely meet irreducible in the congruence lattice. When K is a class of finite algebras, then the congruence lattice is finite, so complete meet irreducibility coincides with meet irreducibility, and so for the purposes of this paper, either notion would suffice.

In general there may not be a unique maximum congruence separating a pair $a \neq b$, but Definition 2.5 only requires that $\pi_{a,b}$ be a maximal congruence separating a from b.

The following gives a one-to-many first order reduction from an algebra with definable cmi congruences to its subdirectly irreducible quotients. This in turn provides a one-to-many first order reduction from $\mathsf{var\text{-}mem}(*,\mathbf{B})$ to $\mathsf{var\text{-}si\text{-}mem}(*,\mathbf{B})$.

Theorem 2.6. *If $\mathcal{K}$ is a class of algebras of the same signature, closed under taking homomorphic images and with definable cmi congruences, then there is a first order query $\phi(x,y)$ with two free variables x,y, such that for every $\mathbf{A} \in \mathcal{K}$ and every $a \neq b$ in A we have $\phi(a,b)$ defines a subdirectly irreducible quotient of $\mathbf{A}$ in which the monolith is generated by representatives of the congruence classes of a and b.*

Proof. Let $\pi_{x,y}(u,v)$ be a formula defining cmi congruences in K, and for each $\mathbf{A} \in K$ and each $a \neq b$ in A, let $\pi_{a,b}$ denote the cmi congruence defined by $\pi_{a,b}(u,v)$. Our one-to-many reduction will use an implicit linear order $\leq$ on the set A (with $<$ being the corresponding strict order).

We first observe that we may use $\leq$ to define the set of minimal representatives of the $\pi_{a,b}$-classes: $\{z \in A \mid \forall x(\pi_{a,b}(x,z) \Rightarrow z \leq x)\}$. On this set we define each fundamental n-ary operation f as follows:

$$(f(c_1,\dots,c_n) \approx d) \leftrightarrow (\pi_{a,b}(f(c_1,\dots,c_n),d)).$$

The operations are well defined because $\pi_{a,b}$ is a congruence relation, and as we have selected the domain as a transversal of the congruence classes, the defined algebra is a subdirectly irreducible quotient of $\mathbf{A}$ in which the representative members of the $\pi_{a,b}$ congruence classes of a and b generate the minimal congruence. □

Note that when there is a semilattice operation $\wedge$, we may avoid recourse to the implicit linear order $\leq$ by instead taking $\wedge$-minimum of each congruence class. We deploy this approach in Section 5.

Corollary 2.7. *If $\mathcal{V}$ has definable cmi congruences, by formula $\pi_{x,y}(u,v)$, then $\mathcal{V}_{\mathrm{si}}$ is definable within $\mathcal{V}$ by the sentence*

$$\exists x \exists y \forall u \forall v \; x \neq y \;\; \& \;\; (u \approx v \leftrightarrow \pi_{x,y}(u,v)).$$

We mention that subdirectly irreducible algebras are also first order definable in the presence of definable principal congruences, as subdirect irreducibility is equivalent to

$$\exists u \exists v \forall x \forall y \; x \not\approx y \rightarrow (u,v) \in \mathrm{Cg}(x,y).$$

To complete the section we give an example property that guarantees definable cmi congruences, and therefore also the identification of subdirectly irreducible quotients as first order queries. For any term $t(x,z_1,\dots)$ in T_x, we define the formula $\rightsquigarrow_t(x,y)$ by

$$\exists z_1 \exists z_2 \dots t(x,z_1,z_2,\dots) \approx y.$$

If $\mathbf{A}$ is an algebra and $a, b \in A$ are such that $\rightsquigarrow_t(a, b)$ holds on $\mathbf{A}$, then we say that a *divides* b by way of t, and write $a \rightsquigarrow_t b$. More generally, for a set of terms $F \subseteq T_x$ we similarly write $\rightsquigarrow_F$ to denote the possibly infinite disjunction $\bigvee_{t \in F} \rightsquigarrow_t$. We say that a divides b to mean that $a \rightsquigarrow_t b$ for some t, or equivalently if $a \rightsquigarrow_{T_x} b$ (restricting to those terms of T_x explicit in x). A class has *finitely determined division* (FDD) if the relation $\rightsquigarrow$ is first order definable, or equivalently if there is a finite set of terms F such that $\rightsquigarrow = \rightsquigarrow_F$ (this equivalence is at the level of arbitrary algebraic structures, not finite algebraic structures; [36, Theorem 5.1]). As is shown in Example 4.3 of [36], the FDD property properly generalises the TFPC property.

The relation $\rightsquigarrow$ is always preorder, but if it is an order relation on some algebra (or class of algebras), then we say that algebra (or class of algebras, respectively) is *division ordered*. Theorem 7.3 of [36] shows that the finite division ordered algebras of a given signature form a pseudovariety[1] and contain a number of finite algebras with negative universal algebraic properties: Lyndon's algebra [48] for example. The restricted case of division ordered semigroups is particularly well studied and known as the $\mathcal{J}$-trivial semigroups (see Exercise 5.1.5 in Almeida [2]) and have also been an important source of challenging examples in semigroup varieties; see Gusev and Sapir [23] for one of many examples.

We have framed the following result in the context of universal Horn classes, because the division order property is defined by quasiequations: $\{x \rightsquigarrow_t y \;\; \& \;\; y \rightsquigarrow_s x \to x \approx y \mid s, t \in T_x\}$. As we shall shortly see (Lemma 3.1(i) below, due to Gorbunov), the result also applies to pseudovarieties of division ordered finite algebras.

Proposition 2.8. *Let $\mathcal{K}$ be a universal Horn class of algebras with TFPC. If $\mathcal{K}$ is division ordered, then $\mathcal{K}$ has definable cmi congruences.*

Proof. First note that Example 4.2 of [36] shows that the TFPC condition ensures that there is a finite set of terms $F \subseteq T_x$ such that $\mathcal{K}$ has TFPC and FDD by way of F. In particular $\rightsquigarrow$ coincides with $\rightsquigarrow_F$ and so is a first order definable relation. We also note that there is no loss of generality in assuming that $x \in F$; this is a particular case of the General Shadowing Theorem 3.1 of [13]. As $\mathcal{K}$ is division ordered, either $a \not\rightsquigarrow b$ or $b \not\rightsquigarrow a$. Assume that $a \not\rightsquigarrow b$ and define $J_b := \{c \mid c \not\rightsquigarrow b\}$, noting that $a \in J_b$ and $b \notin J_b$. As $\rightsquigarrow$ is first order definable in $\mathcal{K}$, so also is the unary relation J_b. Now recall the *syntactic congruence* of $\sim_S$ of a subset $S \subseteq A$, as defined by

$$u \sim_S v \Leftrightarrow \mathop{\&}_{t \in F} \forall \vec{z}\, t(u, \vec{z}) \in S \leftrightarrow t(v, \vec{z}) \in S.$$

The TFPC condition with respect to F ensures this is a congruence, and indeed the largest congruence in which S is a union of congruence classes (see [13, Corollary 2.4] for the specific usage of TFPC, though the underlying idea of syntactic congruence is very well known in semigroup theoretic contexts and dates back at least as far as the 1954 paper of Pierce [54]). Now the relation $\sim_S$ is a first order formula in the unary predicate S. As J_b is a first order definable unary relation, it follows that $\sim_{J_b}$ is a first order definable congruence relation. We now claim that it is a maximal congruence separating a from b. First observe that J_b is a single congruence class of $\sim_{J_b}$. In particular, as $a \in J_b$ we have that $a \not\sim_{J_b} b$. We claim that any congruence θ extending $\sim_{J_b}$ collapses b to some element of J_b. As $\sim_{J_b}$ is the maximum congruence with respect to not collapsing an element from $A \backslash J_b$ to an element of J_b, it follows that there is $b' \in A \backslash J_b$ and $a' \in J_b$ with $a' \;\theta\; b'$. By

[1] Theorem 7.3 of [36] concerns the variety of an individual algebra, but the proof depends only on showing that a quotient of a locally finite division ordered algebra is division ordered, and this is sufficient to deduce that the class of finite division ordered algebras is closed under taking quotients; it is trivially closed under the taking of finitary direct products and subalgebras.

the definition of J_b there is a term $t(x, \vec{z}) \in F$, and elements $\vec{c}$ from A such that $t^{\mathbf{A}}(b', \vec{c}) = b$. Then $t^{\mathbf{A}}(b', \vec{c})$ is congruent to $t^{\mathbf{A}}(a', \vec{c})$ modulo θ. But $t^{\mathbf{A}}(a', \vec{c}) \in J_b$ as $a' \in J_b$ and J_b is an absorbing ideal of $\mathbf{A}$. This shows that the first order relation $\sim_{J_b}$ is a maximal congruence with respect to not identifying a and b; but we assumed $a \not\sim b$. As $\mathcal{K}$ is division ordered, either $a \not\sim b$ or $b \not\sim a$ and so $\mathcal{K}$ has definable cmi congruences by way of the universal formula

$$\pi_{x,y}(u, v) = \big(x \not\sim_F y \ \ \& \ \ u \sim_{J_y} v\big) \vee (y \not\sim_F x \ \ \& \ \ u \sim_{J_x} v)\,. \qquad \square$$

We mention the work of Schein [61], where for example Theorem 3.7 (which concern semigroups with zero for which $\sim_{\{0\}}$ is the identity relation) or Theorems 4.5 and 4.6 (which concern a subclass of the $\mathcal{J}$-trivial semigroups) might serve as a motivation for the proof idea of Proposition 2.8, even if [61] concerns the structure of subdirectly irreducible semigroups rather than the potential definability of subdirectly irreducible quotients.

3. Toolkit: Preservation Theorems

In this section we provide some preservation theorems that hold at the finite level in algebraic signatures. We use some of these to provide information on tightness to the number of quantifier alternations in the results of Section 5.

In the class of relational structures, HSP-classes are too restrictive to be of much interest. However there is a rather elementary preservation theorem in universal first order logic that can be stated for SP-classes for arbitrary finite structures (relational or algebraic), provided uniform local finiteness is assumed. To begin with we recall some basic facts concerning classes generated by classes of finite structures. In the following, $\mathrm{Th}_{\mathrm{uH}}(\mathcal{K})$ denotes the universal Horn theory of $\mathcal{K}$.

Lemma 3.1. *The following hold for any $\mathsf{SP}_{\mathrm{fin}}$-closed class $\mathcal{K}$ of finite structures in a finite signature.*

(1) $\mathcal{K} = \mathrm{Mod}_{\mathrm{fin}}(\mathrm{Th}_{\mathrm{uH}}(\mathcal{K}))$.
(2) *$\mathcal{K}$ is uniformly locally finite if and only if* $\mathrm{Mod}(\mathrm{Th}_{\mathrm{uH}}(\mathcal{K}))$ *is locally finite.*
(3) *$\mathcal{K}$ is uniformly locally finite if and only if the ultraproduct closure of $\mathcal{K}$ is locally finite.*
(4) *If $\mathcal{K}$ is a uniformly locally finite class of algebras and is also closed under taking homomorphic images, then $\mathcal{K}$ is the finite part of* $\mathsf{HSP}(\mathcal{K})$.

Proof. The first part is due to Gorbunov [21, Proposition 2.5.12], while the second part is usually attributed to Maltsev. The "only if" statement of part (3) follows from part (2) because the ultraproduct closure of $\mathcal{K}$ is a subclass of the universal Horn class generated by $\mathcal{K}$. For the "if" statement, we prove the contrapositive. Assume that $\mathcal{K}$ is not uniformly locally finite, so that there exists an $n \in \mathbb{N}$ such that for every $m \in \mathbb{N}$ there is an n-generated member $\mathbf{S}_m$ of $\mathcal{K}$ with more than m elements. Add constants $a_1, \ldots, a_n$ to name the n generators of each $\mathbf{S}_m$. For each $k \in \mathbb{N}$ there are, up to isomorphism, only finitely many $\{a_1, \ldots, a_n\}$-generated structures of size up to k and so we may create a variable-free sentence σ_k (in the expanded signature including the constants $a_1, \ldots, a_n$) that asserts that $\mathbf{S}_m$ is not amongst them, whenever $m \geq k$. For example, σ_k could be the negation of the disjunction of the diagrams of (representatives of the isomorphism classes) of all at most k-element structures generated by $a_1, \ldots, a_k$; see [20, Definition 8.35]. Then the ultraproduct of $\{\mathbf{S}_m \mid m \in \mathbb{N}\}$, over any nonprincipal ultrafilter on $\mathbb{N}$, also satisfies σ_k, and as this is true for every k, it follows that the substructure of the ultraproduct generated by $a_1, \ldots, a_n$ is infinite. So, the ultraproduct closure of $\mathcal{K}$ is not locally finite, as required for the contrapositive of the "if" statement of (3).

For part (4), note that every variety is generated by its finitely generated free algebras, and all finite algebras are quotients of free algebras. Uniform local finiteness ensures that finitely generated free algebras in the variety generated by $\mathcal{K}$ are finite, and as $\mathcal{K}$ is assumed to be closed under quotients, these free algebras, and their quotients, lie in $\mathcal{K}$. In other words, $\mathcal{K}$ is the finite part of $\mathsf{HSP}(\mathcal{K})$. □

The following easy observation gives some partial preservation theorems at the finite level.

Theorem 3.2. *Let Φ be a universal sentence defining a locally finite class.*

(1) If the finite models $\mathcal{K} = \mathrm{Mod}_{\mathrm{fin}}(\Phi)$ are $\mathsf{SP}_{\mathrm{fin}}$-closed then $\mathrm{Mod}(\Phi)$ is a universal Horn class and Φ is logically equivalent to a universal Horn sentence, even amongst structures of possibly infinite size.

(2) If the finite models $\mathcal{K} = \mathrm{Mod}_{\mathrm{fin}}(\Phi)$ are a pseudovariety ($\mathsf{HSP}_{\mathrm{fin}}$-closed) then $\mathrm{Mod}(\Phi)$ is a variety and Φ is logically equivalent to a finite conjunction of equations, even amongst structures of possibly infinite size.

Proof. To prove (1), first observe that as $\mathcal{K} \subseteq \mathrm{Mod}(\Phi)$ and $\mathrm{Mod}(\Phi)$ is locally finite and ultraproduct-closed, it follows from Lemma 3.1 part (3) that $\mathcal{K}$ is uniformly locally finite. Then Lemma 3.1 part (2) implies that $\mathrm{Mod}(\mathrm{Th}_{\mathrm{uH}}(\mathcal{K}))$ is locally finite.

Next observe that two locally finite universal classes agreeing on finite models coincide, because all models embed into an ultraproduct of their finitely generated substructures. So $\mathrm{Mod}(\mathrm{Th}_{\mathrm{uH}}(\mathcal{K})) = \mathrm{Mod}(\Phi)$, and then the Compactness Theorem ensures that there is a finite subset of $\mathrm{Th}_{\mathrm{uH}}(\mathcal{K})$ that is equivalent to Φ.

The proof of (2) is essentially identical, using Lemma 3.1(4) in place of (3). □

The following theorem differs subtly from Theorem 3.2 in that there is no requirement that $\mathrm{Mod}(\Phi)$ be locally finite, only that $\mathrm{Mod}_{\mathrm{fin}}(\Phi)$ be uniformly locally finite. This is a much weaker assumption, and the conclusion is weaker in dropping the conclusion of equivalence on all structures. We phrase it for algebraic signatures, though it can be adapted for general signatures.

Theorem 3.3. *Let Φ be a universal sentence in an algebraic signature whose finite models $\mathcal{K} = \mathrm{Mod}_{\mathrm{fin}}(\Phi)$ are $\mathsf{SP}^{+}_{\mathrm{fin}}$-closed and uniformly locally finite. Then Φ is equivalent on finite algebras to a universal Horn sentence, and if $\mathcal{K}$ is a pseudovariety, then Φ is equivalent on finite structures to a finite conjunction of equations.*

Proof. Let n be the number of variables appearing in Φ. Birkhoff's Finite Basis Theorem 1.5 guarantees that the n-variable equational theory of $\mathcal{K}$ has a finite basis Σ_{BFB}, and all n-variable terms reduce modulo Σ_{BFB} to one in the finite set T_{BFB}. Up to logical equivalence, there are only finitely many n-variable atomic formulæ and only finitely many universal Horn sentences involving only terms from T_{BFB}. Let Σ be the conjunction of Σ_{BFB} along with the finitely many satisfied universal Horn sentences involving only terms T_{BFB} (and when $\mathcal{K}$ is a pseudovariety, it will suffice to take only Σ_{BFB}). The universal Horn sentence Σ determines when an n-generated finite algebra lies in $\mathcal{K}$. But as $\mathcal{K} = \mathrm{Mod}_{\mathrm{fin}}(\Phi)$ and Φ is an n-variable universal sentence, membership of a finite algebra $\mathbf{A}$ in $\mathcal{K}$ is equivalent to membership in $\mathcal{K}$ of the (at most) n-generated subalgebras of $\mathbf{A}$. Thus Σ is equivalent to Φ on finite algebras. □

The following result is given for function-free signatures in [22, Theorem 3], where it is attributed to Compton. Recall that a *hereditary class* means one that is closed under S. For an algebra $\mathbf{A}$ and subset $B \subseteq A$ we let $\mathrm{sg}_{\mathbf{A}}(B)$ denote the subalgebra of $\mathbf{A}$ generated by B.

Theorem 3.4. *If Φ is a $\forall^*\exists^*$-sentence defining a uniformly locally finite hereditary class $\mathcal{K}$ of finite structures then Φ is equivalent on finite structures to a universal sentence.*

Proof. The proof is by way of a sort of "localised Skolemisation" process. Let Φ be

$$\forall x_1 \forall x_2 \dots \forall x_n \exists y_1 \exists y_2 \dots \exists y_m \phi(x_1, \dots, x_n, y_1, \dots, y_m).$$

We first observe that uniform local finiteness allows us to apply Birkhoff's Finite Basis Theorem 1.5 to find a finite set F of terms in variables $x_1, \dots, x_n$ such that every term $t(x_1, \dots, x_n)$ is equivalent in $\mathcal{K}$ to one from F; that is, for every $t(x_1, \dots, x_n)$ there exists $s \in F$ with $\mathcal{K} \models s \approx t$.

We claim that Φ is equivalent to the sentence $\Phi' = \forall x_1 \dots \forall x_n \phi'(x_1, \dots, x_n)$ where

$$\phi'(x_1, \dots, x_n) \coloneqq \bigvee_{(t_1,\dots,t_m) \in F^m} \phi(x_1, \dots, x_n, t_1(x_1, \dots, x_n), \dots, t_m(x_1, \dots, x_n)).$$

It is trivial that $\Phi' \vdash \Phi$, so that the finite models of Φ' are a subclass of $\mathcal{K}$. Now assume that $\mathbf{A} \in \mathcal{K}$. Consider any $a_1, \dots, a_n \in A$. Now as $\mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n) \models \Phi$, there are $b_1, \dots, b_m \in \mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n)$ with $\mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n) \models \phi(a_1, \dots, a_n, b_1, \dots, b_m)$. As $b_1, \dots, b_m \in \mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n)$ there are terms $t_1, \dots, t_m \in F$ with $b_i = t_i^{\mathbf{A}}(a_1, \dots, a_n)$ for $i = 1, \dots, m$. Then

$$\mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n) \models \phi(a_1, \dots, a_n, t_1^{\mathbf{A}}(a_1, \dots, a_n), \dots, t_m^{\mathbf{A}}(a_1, \dots, a_n)),$$

and hence $\mathrm{sg}_{\mathbf{A}}(a_1, \dots, a_n) \models \phi'(a_1, \dots, a_n)$. That is, $\mathbf{A} \models \phi'(a_1, \dots, a_n)$. As $a_1, \dots, a_n$ were arbitrary, it follows that $\mathbf{A} \models \Phi'$. □

The following result also holds if "pseudovariety" is replace by "$\mathsf{SP}^+_{\mathrm{fin}}$-closed class", and "finite conjunction of equations" is replaced by "universal Horn sentence".

Corollary 3.5. *If Φ is a $\forall^*\exists^*$ sentence defining a uniformly locally finite pseudovariety $\mathcal{K}$ of finite algebras, then Φ is equivalent on finite structures to a finite conjunction of equations.*

Proof. Theorem 3.4 shows that Φ is equivalent to a universal sentence, and then Theorem 3.3 shows this is equivalent to a conjunction of equations. □

Another way to state Corollary 3.5 is that a uniformly locally finite pseudovariety $\mathcal{K}$ of finite algebras is either finitely axiomatised by equations, or cannot be defined by any $\forall^*\exists^*$-sentence.

We now observe that the assumption of uniform local finiteness in these statements is necessary (a similar example can be found in [58]).

Example 3.6. *The class of finite $\mathcal{J}$-trivial monoids is definable by a universal Horn sentence, but not by any system of equations. So the Preservation Theorem for HSP-classes fails at the finite level, even amongst universal Horn sentences. The example also shows that Theorems 3.2, 3.3 and Corollary 3.5 do not hold if the assumption of uniform local finiteness is dropped.*

Proof. Our definition of $\mathcal{J}$-triviality in Example 1.2 is by way of a universal Horn sentence. However as all finite nilpotent monoids are $\mathcal{J}$-trivial, all equations true of $\mathcal{J}$-trivial monoids are true of all monoids. So the class of finite $\mathcal{J}$-trivial monoids is not the set of finite models of any system of equations (and certainly not of any finite conjunction of equations). □

4. Toolkit: Ehrenfeucht Fraïssé games in finite algebras

In this section we consider Ehrenfeucht-Fraïssé games on finite algebras. While the reader can find a deeper development of Ehrenfeucht-Fraïssé games in almost any model theory or finite model theory text, the restriction to finite algebras offers some modest improvements that do not seem to be exposited in such texts. Libkin [47] for example develops these games in the finite setting, but restricts to relational signatures (possibly with constants). Hodges [28] presents Ehrenfeucht-Fraïssé games for arbitrary signatures (on not-necessarily finite models), but two versions of the game $EF_k(\mathbf{A}, \mathbf{B})$ and $EF_k[\mathbf{A}, \mathbf{B}]$ (coinciding in relational signatures) are introduced to capture different features, and only un-nested first order formulæ are captured. The present section serves as both a preliminary introduction for later use, particularly in results Section 8, and to make the observation that *finite* algebraic signatures offer a uniform treatment of both algebraic and relational signatures (essentially by way of $EF_k(\mathbf{A}, \mathbf{B})$). For example, we prove completeness of the $EF_k(\mathbf{A}, \mathbf{B})$ game when it comes to establishing the failure of first order definability amongst finite algebras within uniformly locally finite classes.

The board of the game consists two structures $\mathbf{A}$ and $\mathbf{B}$, which in this paper will be finite structures in the same arbitrary finite signature (typically algebraic for us, though not necessarily). The goal of Spoiler is to reveal the two structures as distinct; the goal of Duplicator is to hide any distinction between the two for as long as possible!

The players play a certain number of rounds. Each round consists of the following rules.

(1) Spoiler picks a structure; either $\mathbf{A}$ or $\mathbf{B}$.
(2) Spoiler makes a move by selecting an element of that structure; either $a \in \mathbf{A}$ or $b \in \mathbf{B}$.
(3) Duplicator responds by selecting an element in the other structure.

After n rounds of an Ehrenfeucht-Fraïssé game, we have moves $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$. Duplicator wins the n-round game provided the map that sends each a_i to b_i extends to an isomorphism between the substructures of $\mathbf{A}$ and $\mathbf{B}$ generated by $\{a_1, \ldots, a_n\}$ and $\{b_1, \ldots, b_n\}$ respectively. If Duplicator can win the n-round game regardless of how Spoiler plays, we say that Duplicator has *an n-round winning strategy* and we write $\mathbf{A} \equiv_n \mathbf{B}$. In the conventional setting, where positive arity operations are disallowed, there is almost no generation power: it simply means the usual automatic inclusion of any constants in the signature.

The *quantifier rank* qr of a formula is defined inductively as follows:

- If ϕ is an atomic formula then $\mathrm{qr}(\phi) = 0$;
- $\mathrm{qr}(\phi_1 \vee \phi_2) = \mathrm{qr}(\phi_1 \ \& \ \phi_2) = \max\{\mathrm{qr}(\phi_1), \mathrm{qr}(\phi_2)\}$;
- $\mathrm{qr}(\neg\phi) = \mathrm{qr}(\phi)$;
- $\mathrm{qr}(\forall x\, \phi) = \mathrm{qr}(\exists x\, \phi) = \mathrm{qr}(\phi) + 1$.

We use the notation FO$[n]$ for all first order formulæ of quantifier rank up to n, and in general use the abbreviation FO for *first order logic* (recalling that `FO` refers to the class of first order definable decision problems).

The first fundamental theorem for Ehrenfeucht-Fraïssé games in finite model theory is the following; there is no requirement for the signature to be relational here.

Theorem 4.1. *The following are equivalent for two finite structures $\mathbf{A}$ and $\mathbf{B}$ of the same signature*:

(1) $\mathbf{A} \equiv_n \mathbf{B}$; *that is, Duplicator has a winning strategy in an n-round play of our new game.*
(2) $\mathbf{A}$ *and* $\mathbf{B}$ *agree on* FO[n].

Theorem 4.1 gives rise to a general methodology for proving inexpressibility results, which we give in the next theorem. The theorem is usually restricted to relational signatures (no operations of positive arity) where uniform local finiteness holds always, and so the entire statement can be given as an if and only if. The proof under uniform local finiteness assumptions is the main new contribution in the section, though the arguments are mostly almost identical, but incorporating Birkhoff's Finite Basis Theorem 1.5.

Theorem 4.2. *A property $\mathcal{P}$ is not expressible in FO if for every $n \in \mathbb{N}$, there exist two finite structures $\mathbf{A}_n$ and $\mathbf{B}_n$, such that*:

- $\mathbf{A}_n \equiv_n \mathbf{B}_n$.
- $\mathbf{A}_n$ *has property $\mathcal{P}$, and $\mathbf{B}_n$ does not.*

When relativised to any uniformly locally finite class of structures, the converse also holds: *inexpressibility in FO ensures the existence of $\mathbf{A}_n$ and $\mathbf{B}_n$.*

The following example demonstrates the necessity of some caveat for the "only if" part of Theorem 4.2, and therefore a fundamental difference between relational signatures and algebraic signatures.

Example 4.3. *Consider the signature of* unars*—unary algebras with a single unary operation f—and consider the property $\mathcal{P}$ that $f^n(x) \approx f^{n+1}(x)$ holds for sufficiently large n. Then $\mathcal{P}$ is not first order definable at the finite level, yet unars with property $\mathcal{P}$ can be distinguished from unars failing $\mathcal{P}$ in a one step algebraic Ehrenfreucht-Fraïssé game.*

Proof. Assume $\mathbf{A}$ satisfies $\mathcal{P}$ and $\mathbf{B}$ does not. So there is $b \in B$ such that $f^n(b)$ never coincides with $f^{n+1}(b)$, for any n. Spoiler chooses this b, and for any $a \in A$ that duplicator selects we find that $\mathrm{sg}_{\mathbf{A}}(a) \ncong \mathrm{sg}_{\mathbf{B}}(b)$ because for some n the property $f^n(a) = f^{n+1}(a)$ holds. So $\mathbf{A} \not\equiv_1 \mathbf{B}$.

Now to show that $\mathcal{P}$ is not first order definable at the finite level. For this it suffices to work in the relational setting, where f is replaced by its graph: this is because if there is a first order formula defining $\mathcal{P}$ in the algebraic setting, then it may be translated into one defining $\mathcal{P}$ in the relational setting. Let $\mathbb{A}_k$ denote the digraph consisting of a path of length k with a loop as terminal vertex, and let $\mathbb{B}_k$ denote the digraph consisting of the disjoint union of $\mathbb{A}_k$ with a k-cycle. It is routine exercise in standard Ehrenfreucht-Fraïssé games to verify that for sufficiently large k, Duplicator may survive n rounds of this game; see Figure 3.4 of Libkin [47] and accompanying text for example. □

Define the *height* of a term as follows: variables are of height 0. If f is a fundamental operation of arity n and the maximum height amongst some set of terms $t_1, \ldots, t_n$ is k, then $f(t_1, \ldots, t_n)$ is of height $k + 1$. We interpret this as defining the height of nullaries to be 1.

The following lemma is an algebraic variant of Lemma 3.13 of Libkin [47].

Lemma 4.4. *Let k, m, n be natural numbers.*

(1) *Every atomic subformula of a rank k formula in m free variables $x_0, \ldots, x_{m-1}$ involves at most $m + k$ variables.*
(2) *Up to logical equivalence there are only finitely many distinct rank k formulæ in the m free variables $x_0, \ldots, x_{m-1}$ whose terms are of height at most n.*

Proof. Both statements are proved by induction over k (the number m varies: for each k we ask the statement be true of any m). We omit full details of the proof as it is essentially identical to as in [47, Lemma 3.13], with the one difference occurring in the second item, in consideration of the base case of a rank 0 formula. However the difference is minor: the bound on height of terms continues to ensure that there are only finitely many atomic formulæ in m free variables, and every rank 0 formula is a Boolean combination of these, and hence the base statement is true. □

Definition 4.5. Let $\mathbf{A}$ be a finite algebra and $a \in A$. Define the *rank k type* of the element a, denoted $\mathrm{tp}_k(\mathbf{A}, a)$, to be the set of all FO-formulæ $\varphi(x)$ of rank k for which $\mathbf{A} \models \varphi(a)$.

The set $\mathrm{tp}_k(\mathbf{A}, a)$ is always infinite because there is no bound on height of any terms that appear within. The following lemma contains the primary use of uniform local finiteness.

Lemma 4.6. *Let $\mathbf{A}$ be a finite algebra and $a \in A$. Then the rank k type $\mathrm{tp}_k(\mathbf{A}, a)$ of a is logically equivalent to a finite subset of $\mathrm{tp}_k(\mathbf{A}, a)$.*

Proof. We use the notation and statement of Birkhoff's Finite Basis Theorem 1.5, specifically the sets $T_{\mathrm{BFB}}(\mathsf{HSP}_{\mathrm{fin}}(\mathbf{A}), k+1)$ and $\Sigma_{\mathrm{BFB}}(\mathsf{HSP}_{\mathrm{fin}}(\mathbf{A}), k+1)$, noting that we are using $k+1$ rather than k as in the statement. Each equation

$$f(u_{i_1}(x_0, x_1, \ldots, x_k), \ldots, u_{i_n}(x_0, x_1, \ldots, x_k)) \approx u_{i'}(x_0, x_1, \ldots, x_k)$$

in Σ_{BFB} is satisfied universally by $\mathbf{A}$, and hence the formula

$$\forall x_1 \ldots \forall x_k \; f(u_{i_1}(x, x_1, \ldots, x_k), \ldots, u_{i_n}(x, x_1, \ldots, x_k)) \approx u_{i'}(x, x_1, \ldots, x_k)$$

is a rank k formula satisfied at $x = a$, and so is contained in the set $\mathrm{tp}_k(\mathbf{A}, a)$. Let Σ'_{BFB} be the set of these formulæ: the equations in Σ_{BFB} but with the first variable listed as x and not quantified. Let $\Psi(x)$ be an arbitrary formula in $\mathrm{tp}_k(\mathbf{A}, a)$, and let $s(x, x_1, \ldots, x_k) \approx t(x, x_1, \ldots, x_k)$ be an atomic subformula of $\Psi(x)$, noting that the first statement in Lemma 4.4 implies that there are indeed (at most) $k+1$ distinct variables. Then by Birkhoff's Finite Basis Theorem 1.5 part (3), there are i, j such that $\mathbf{A} \models s(x, x_1, \ldots, x_k) \approx u_i(x, x_1, \ldots, x_{i_k})$ and $\mathbf{A} \models t(x, x_1, \ldots, x_m) \approx u_i(x, x_1, \ldots, x_{i_k})$, and by part (4), there is a proof of these equalities that proceeds by direct replacement using laws in Σ_{BFB}. Thus we can replace the atomic subformula $s(x, x_1, \ldots, x_k) \approx t(x, x_1, \ldots, x_k)$ in Ψ by $u_i(x, x_1, \ldots, x_{i_k}) \approx u_j(x, x_{i_1}, \ldots, x_{i_k})$, where direct replacement is used because the value of x is to be fixed as the single element a so cannot be substituted. Thus $\Psi(x)$ is equivalent to one in which all terms have height at most the maximum height of the terms in $T_{\mathbf{BFB}}$. Thus, by the second part of Lemma 4.4, the set $\mathrm{tp}_k(\mathbf{A}, a)$ is logically equivalent to the finite subset consisting of Σ'_{BFB} along with all members of $\mathrm{tp}_k(\mathbf{A}, a)$ whose terms are of height at most n. □

The following is essentially Theorem 3.15 of Libkin [47] in the algebraic setting. As the (easy) proof is unchanged we omit it, though comment that the sentence $\alpha_k(x)$ is of course just the conjunction of the members of the finite subset of $\mathrm{tp}_k(\mathbf{A}, a)$ identified as equivalent to $\mathrm{tp}_k(\mathbf{A}, a)$ in Lemma 4.6.

Lemma 4.7. *For any finite algebra $\mathbf{A}$ and element $a \in A$ there is a rank k formula $\alpha_k(x)$ such that $\mathbf{A} \models \alpha_k(a)$ and any similar algebra $\mathbf{B}$ with $b \in B$ satisfying $\alpha_k(b)$ has $\mathrm{tp}_k(\mathbf{B}, b) = \mathrm{tp}_k(\mathbf{A}, a)$.*

We shall prove the equivalence of (1) and (2) in Theorem 4.1, as well as the following the back-and-forth equivalence (a standard step in this argument, define here for convenience).

Definition 4.8. We inductively define the back-and-forth relations $\simeq_k$ on finite $\mathcal{F}$-structures $\mathbf{A}$ and $\mathbf{B}$ as follows.

- $\mathbf{A} \simeq_0 \mathbf{B}$ iff $\mathbf{A} \equiv_0 \mathbf{B}$; that is, $\mathbf{A}$ and $\mathbf{B}$ satisfy the same atomic sentences.
- $\mathbf{A} \simeq_{k+1} \mathbf{B}$ iff the following two conditions hold:
  forth: for every $a \in \mathbf{A}$, there exists $b \in \mathbf{B}$ such that $(\mathbf{A}, a) \simeq_k (\mathbf{B}, b)$.
  back: for every $b \in \mathbf{B}$, there exists $a \in \mathbf{A}$ such that $(\mathbf{A}, a) \simeq_k (\mathbf{B}, b)$.

The following extension of Theorem 4.1 now follows using the standard proof.

Theorem 4.9. *Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be two finite structures of the same signature. Then the following are equivalent:*

(1) $\mathbf{A}$ *and* $\mathbf{B}$ *agree on FO[k];*
(2) $\mathbf{A} \equiv_k \mathbf{B}$;
(3) $\mathbf{A} \simeq_k \mathbf{B}$.

Proof. We omit the proof, as the one given by Libkin for [47, Theorem 3.18] (for example) holds close to verbatim, with the only adjustments for algebraic setting having occurred in the adjusted proof of Lemma 4.7 and its corollary Lemma 4.6 (which are used in the proof of (1) $\implies$ (3)), and in the meaning of the relations $\equiv_k$, $\simeq_k$. We note that the primary difference in the meaning of $\equiv_k$, $\simeq_k$ manifests at the trivial base step: a statement

$$(\mathbf{A}, a_1, \ldots, a_n) \simeq_0 (\mathbf{B}, b_1, \ldots, b_n)$$

(which is the base $\simeq_0$ assumption founding the relation $\simeq_n$) is equivalent to the subalgebra $\mathrm{sg}_{\mathbf{A}}(a_1, \ldots, a_n)$ being isomorphic to $\mathrm{sg}_{\mathbf{B}}(b_1, \ldots, b_n)$ under the map sending each a_i to the corresponding b_i. But this matches our game-based definition of $\equiv_0$ and of agreement on FO[0]. □

Finally, we prove the final statement in Theorem 4.2, showing completeness of the method for detecting failure of first order definability within uniformly locally finite classes of finite algebras. We do this by proving the contrapositive: a property $\mathcal{P}$ of a uniformly locally finite class of finite algebras $\mathcal{U}$ is expressible in FO iff there exists a number k such that for every two algebras $\mathbf{A}$ and $\mathbf{B}$ in $\mathcal{U}$, if $\mathbf{A} \in \mathcal{P}$ and $\mathbf{A} \equiv_k \mathbf{B}$, then $\mathbf{B} \in \mathcal{P}$.

Proof of final statement in Theorem 4.2. Let $\mathbf{A}$ and $\mathbf{B}$ be in the uniformly locally finite $\mathcal{U}$. If $\mathcal{P}$ is expressible by a FO sentence Φ, let $k := \mathrm{qr}(\Phi)$. If $\mathbf{A} \in \mathcal{P}$, then $\mathbf{A} \models \Phi$, and hence for $\mathbf{B}$ with $\mathbf{A} \equiv_k \mathbf{B}$ we have $\mathbf{B} \models \Phi$. Thus $\mathbf{B} \in \mathcal{P}$.

Conversely, if $\mathbf{A} \in \mathcal{P}$ and $\mathbf{A} \equiv_k \mathbf{B}$ imply $\mathbf{B} \in \mathcal{P}$, then any two algebras in $\mathcal{U}$ with the same rank-k type agree on $\mathcal{P}$. Now, as $\mathcal{U}$ is a uniformly locally finite class of finite algebras, there is a bound on the size of the n-generated algebras and so there are only finitely many different rank-k types. Thus, for every pair of algebras in $\mathcal{U}$ with the same rank-k type, there exists a rank-k formula, namely $\alpha_k(x)$ defined in Lemma 4.7, such that $\mathbf{A} \models \alpha_k(a)$ iff $\mathbf{B} \models \alpha_k(b)$. Hence $\mathcal{P}$ is just a disjunction of some of the α_k's (those witnessed by members of $\mathcal{U}$). □

To conclude this section, we mention a further useful trick for demonstrating non first order definability, this time calling on the classical model theoretic ultraproduct. This simple but effective idea has been used recently by Czédli [16, §2] for example; we give the proof for completeness.

Lemma 4.10. *Let* $\mathcal{K}$ *be a class of finite structures such that for all large enough* n *there are finite structures* $\mathbf{A}_n \in \mathcal{K}$ *and* $\mathbf{B}_n \notin \mathcal{K}$ *for which there is an ultraproduct* $\prod_n \mathbf{A}_n/\mathcal{U} \cong \prod_n \mathbf{B}_n/\mathcal{U}$. *Then* $\mathcal{K}$ *is not first order definable at the finite level.*

Proof. Any first order sentence Φ true of $\mathcal{K}$ is true of all of the $\mathbf{A}_n$, and then of $\prod_{\mathbb{N}/\mathcal{U}} \mathbf{A}_n$, and therefore of cofinitely many of the $\mathbf{B}_n$. Hence Φ does not define $\mathcal{K}$. □

An example usage of this lemma will be given in Proposition 7.3 below.

5. A first order definable pseudovariety that is not definable by a $\forall^*\exists^*$ sentence.

The construction of a nonfinitely based finite algebra with variety membership problem in FO will be based on an idea used in Clark, Davey, Jackson and Pitkethly [14, Theorem 4.2]: the congruence classes of any congruence on a finite lattice are intervals. The result in [14] extends beyond finite algebras to infinite Boolean topological algebras, but in the restriction to finite structures, it suffices to have just a semilattice $\wedge$, as we now prove.

Theorem 5.1. *If $\{\mathbf{L}_1, \ldots, \mathbf{L}_k\}$ is a finite set of semilattice-based finite algebras, then the class $\mathsf{SP}^+_{\mathrm{fin}}(\{\mathbf{L}_1, \ldots, \mathbf{L}_k\})$ is the class of finite models of a $\forall^*\exists^*\forall^*$ sentence in first order logic. The same is true for $\mathsf{SP}_{\mathrm{fin}}(\{\mathbf{L}_1, \ldots, \mathbf{L}_k\})$.*

Before proving Theorem 5.1 we prove a lemma. With minor modifications, this lemma (and then Theorem 5.1) can be adapted to include relations.

Lemma 5.2. *Let $\mathcal{L}$ be the class of finite algebras in some finite signature including a binary operation $\wedge$ with respect to which, all members of $\mathcal{L}$ are semilattices. Fix any $\mathbf{L} \in \mathcal{L}$. Then there is*

- *a first order $\exists^*\forall^*$-sentence $\Xi_{\mathbf{L}}$ asserting the existence of a surjective homomorphism onto $\mathbf{L}$,*
- *a first order $\exists^*\forall^*$-formula $\not\equiv_{\mathbf{L}}(x, y)$ (written $x \not\equiv_{\mathbf{L}} y$) in two free variables such that for every member $\mathbf{K} \in \mathcal{L}$ and every pair $a \neq b$ in K, we have that there is a homomorphism ϕ from $\mathbf{K}$ onto $\mathbf{L}$ with $\phi(a) \neq \phi(b)$ if and only if $\mathbf{K} \models a \not\equiv_{\mathbf{L}} b$.*

Proof. This is essentially a version of the first part of the proof of [14, Theorem 4.2]. Let the universe of $\mathbf{L}$ be $\{1, 2, \ldots, \ell\}$. When there is a surjective homomorphism $\phi : \mathbf{K} \to \mathbf{L}$, then the kernel of ϕ can be understood entirely using the minimal elements of each congruence class: two elements a, b are equivalent modulo this kernel if and only if they sit above precisely the same minimal elements: $\&_{1 \leq i \leq \ell}(a \geq x_i \leftrightarrow b \geq x_i)$, where the x_i are variables that will later be existentially quantified over some larger expression; on satisfaction of this expression, the x_i variables will eventually evaluate as minimal elements in congruence classes. We denote this relation by $\equiv_L$ (without boldface), noting that it is clearly an equivalence relation subject to any evaluation of the variables $x_1, \ldots, x_\ell$. Notationally we write $a \equiv_L b$. The fact that the kernel is a congruence on $\mathbf{K}$ can be asserted by the property that for each operation f, of arity n say, we have

$$\forall a_1, \ldots, a_n, b_1, \ldots, b_n \left(\mathop{\&}_{j=1}^{n} a_j \equiv_L b_j \right) \to f(a_1, \ldots, a_n) \equiv_L f(b_1, \ldots, b_n)$$

The fact that ϕ maps onto $\mathbf{L}$ can also be asserted using $\equiv_L$: for each fundamental operation f (again, of arity n, say), and for each value $f(i_1, \ldots, i_n) = j$ in $\mathbf{L}$ we require $f(x_{i_1}, \ldots, x_{i_n}) \equiv_L x_j$; now take the conjunction over $j = 1, \ldots, \ell$ and existentially quantify $x_1, \ldots, x_\ell$. We have created the $\exists^*\forall^*$ sentence $\Xi_{\mathbf{L}}$ asserting that there is a surjective homomorphism from $\mathbf{K}$ onto $\mathbf{L}$. We wished to define the corresponding relation $\not\equiv_{\mathbf{L}}$ (now with boldface and negated). To do this, we introduce two free variables x, y and include $\neg(x \equiv_L y)$ as a further conjunct within $\Xi_{\mathbf{L}}$. □

Now we may prove Theorem 5.1.

Proof of Theorem 5.1. An algebra $\mathbf{A}$ is in the universal Horn class of $\{\mathbf{L}_1, \dots, \mathbf{L}_k\}$ if and only if there is at least one homomorphism from $\mathbf{A}$ to one of the $\mathbf{L}_i$ and the following "separation properties" hold: whenever $a, b \in A$ have $a \neq b$, there is an $i \in \{1, \dots, k\}$ and homomorphism from $\mathbf{A}$ to $\mathbf{L}_i$ with $\phi(a) \neq \phi(b)$. As $\{\mathbf{L}_1, \dots, \mathbf{L}_k\}$ is a finite set of finite algebras, we may enumerate the set of all subalgebras $\mathbf{L}'_1, \dots, \mathbf{L}'_m$. The following sentence (referring to the sentences defined in Lemma 5.2) asserts the required properties:

$$\Big(\bigvee_{1 \leq i \leq m} \Xi_{\mathbf{L}'_i} \Big) \ \& \ \forall a \forall b \Big(a \not\approx b \to \bigvee_{1 \leq i \leq m} a \not\equiv_{\mathbf{L}'_i} b \Big)$$

Because Ξ and $\not\equiv$ are $\exists^* \forall^*$, this disjunction is a $\forall^* \exists^* \forall^*$-sentence. Once the semilattice axioms for $\wedge$ have been incorporated, we have a single sentence defining $\mathsf{SP}^+_{\mathrm{fin}}(\{\mathbf{L}_1, \dots, \mathbf{L}_k\})$. The sentence for $\mathsf{SP}_{\mathrm{fin}}(\{\mathbf{L}_1, \dots, \mathbf{L}_k\})$ is the same except that we do not include the $\bigvee_{1 \leq i \leq k} \Xi_{\mathbf{L}'_i}$ conjunct. □

As a corollary of Theorem 5.1 we prove that the first order JP Problem is true for algebras with a semilattice operation; this follows already from Willard's Finite Basis Theorem [67], but the relative simplicity of our proof is suggestive of the possibility of more accessible results of this kind in the finite realm.

References

[1] N. Alechina and Yu. Gurevich, Syntax vs. semantics on finite structures, in *Structures in Logic and Computer Science. A Selection of Essays in Honor of A. Ehrenfeucht*, Springer, 1997, pp. 14–33.

[2] J. Almeida, Finite Semigroups and Universal Algebra, World Scientific 1994.

[3] K.A. Baker, G.F. McNulty and J. Wang, An extension of Willard's Finite Basis Theorem: Congruence meet-semidistributive varieties of finite critical depth, Algebra Universalis 52 (2005), 289–302.

[4] V.P. Belkin, Quasi-identities of finite rings and lattices, Algebra Logika 17 (1978) 247–259 (in Russian).

[5] C. Bergman, Universal Algebra: Fundamentals and Selected Topics, Chapman Hall/CRC Press, 2011.

[6] C. Bergman and G. Slutzki, Complexity of some problems concerning varieties and quasivarieties of algebras, SIAM J. Comput. 30 (2000), 359–382.

[7] C. Bergman and G. Slutzki, Computational complexity of some problems involving congruences on algebras. 15th Annual IEEE Symposium on Logic in Computer Science (Santa Barbara, CA, 2000), 168–174, IEEE Comput. Soc. Press, Los Alamitos, CA, 2000.

[8] C. Bergman and G. Slutzki, Computational complexity of some problems involving congruences on algebras, Theoret. Comp. Sci. 270 (2002), 591–608.

[9] G. Bezhanishvili, Locally finite varieties, Algebra Universalis 46 (2001), 531–548.

[10] G. Birkhoff, On the structure of abstract algebras, Math. Proc. Cambridge Phil. Soc. 31 (1935), 433–454.

[11] J. Bulín, D. Delic, M. Jackson and T Niven, A finer reduction of constraint problems to digraphs, Log. Methods Comp. Sci. 11 (2015), 1–33.

[12] S. Burris and H.P. Sankappanavar, A Course in Universal Algebra, Graduate Texts in Mathematics 78, Springer Verlag, 1980.

[13] D.M. Clark, B.A. Davey, R.S. Freese, M. Jackson, Standard topological algebras: Syntactic and principal congruences and profiniteness, Algebra Universalis 52 (2004) 343–376.

[14] D.M. Clark, B.A. Davey, M.G. Jackson and J.G. Pitkethly, The axiomatizability of topological prevarieties, Adv. Math. 218 (2008), 1604–1653.

[15] Clifford A.H. and G.B. Preston, The algebraic theory of semigroups, Vol. I, AMS Mathematical Surveys, (1961).

[16] G. Czédli, Cyclic congruences of slim semimodular lattices and non-finite axiomatizability of some finite structures, Arch. Math. (Brno) 58 (2022), 15–33.

[17] J. Demel, M. Demlová and V. Koubek, Fast algorithms constructing minimal subalgebras, congruences and ideals in a finite algebra, Theoret. Comp. Sci. 36 (1985), 203–216.

[18] S. Eilenberg, Automata languages and machines, Pure and Applied Mathematics, Volume B (Academic Press, New York, 1976).
[19] S. Eilenberg, M.-P. Schützenberger, On pseudovarieties, Adv. Math. 19 (1976), 413–418.
[20] R.S. Freese, R.N. McKenzie, G.F. McNulty and W.F. Taylor, Algebras, Lattices and Varieties, Volume II, Mathematical Surveys and Monographs 268, AMS 2022.
[21] V.A. Gorbunov, Algebraic Theory of Quasivarieties, Consultants Bureau, New York, 1998.
[22] Y. Gurevich, Toward logic tailored for computational complexity, in Computation and Proof Theory, Lecture Notes in Mathematics 1104, 1984, pp. 175–216.
[23] S.V. Gusev and O.B. Sapir, Classification of limit varieties of $\mathcal{J}$-trivial monoids, Comm. Algebra 50 (2002), 3007–3027.
[24] L. Ham, Relativised Homomorphism Preservation at the Finite Level, Studia Logica 105 (2017), 761–786.
[25] P.M. Higgins and M. Jackson, Algebras defined by equations, J. Algebra 555 (2020), 131–151.
[26] P.M. Higgins and M. Jackson, Equationally defined classes of semigroups, Semigroup Forum 107 (2023), 459–477.
[27] D. Hobby and R. McKenzie, The Structure of Finite Algebras, Contemporary Mathematics 76, American Mathematical Society, Providence, RI, 1988.
[28] W. Hodges, Model Theory, Cambridge University Press, 1992.
[29] N. Immerman, Descriptive Complexity, Graduate Texts in Computer Science, Springer 1999.
[30] M. Jackson, Flat algebras and the translation of universal Horn logic to equational logic, J. Symbolic Logic 73 (2008), 90–128.
[31] M. Jackson, Flexible constraint satisfaction and a problem in semigroup theory, Internat. J. Algebra Comput. 35 (2025), 771–821.
[32] M. Jackson, T. Kowalski and T. Niven, Complexity and polymorphisms for digraph constraint problems under some basic constructions, Internat. J. Algebra Comput. 26 (2016), 1395–1433.
[33] M. Jackson and R.N. McKenzie, Interpreting graph colorability in finite semigroups, Internat. J. Algebra Comput. 16 (2006), 119–140.
[34] M. Jackson and G.F. McNulty, The equational complexity of Lyndon's algebra, Algebra Universalis 65 (2011), 243–262.
[35] M. Jackson, M. Ren and X. Zhao, Nonfinitely based ai-semirings with finitely based semigroup reducts, J. Algebra 611 (2022), 211–245.
[36] M. Jackson and B. Trotta, The division relation: congruence conditions and axiomatisability, Comm. Algebra 38 (2010), 534–566.
[37] M. Jackson and B. Trotta, Constraint satisfaction, irredundant axiomatisability and continuous colouring, Studia Logica 101 (2013), 65–94.
[38] M. Jackson and M.V. Volkov, The Algebra of Adjacency Patterns: Rees Matrix Semigroups with Reversion, in A. Blass, N. Dershowitz, and W. Reisig (Eds.): Gurevich Festschrift, LNCS 6300, pp. 414–443, 2010.
[39] H.J. Keisler, Reduced products and Horn classes, Trans. Amer. Math. Soc. 117 (1965) 307–328.
[40] A.Kelarev, Graph Algebras and Automata, CRC Press, Boca Raton, 2003.
[41] O.G. Kharlampovich and M.V. Sapir, Algorithmic problems in varieties, Internat. J. Algebra Comput. 5 (1995), 379–602.
[42] O. Klíma, M. Kunc and L. Polák, Deciding piecewise k-testability, manuscript (2014).
[43] M. Kozik, Computationally and algebraically complex finite algebra membership problems, Internat. J. Algebra Comput. 17 (2007), 1635–1666.
[44] M. Kozik, A 2EXPTIME complete varietal membership problem, SIAM J. Comput. 38 (2009), 2443–2467.
[45] A. Krokhin and S. Živný (eds), The Constraint Satisfaction Problem: Complexity and Approximability, Dagstuhl Follow Ups 7 (2017).
[46] E.W.H. Lee and W.T. Zhang, Finite basis problem for semigroups of order six, LMS J. Comput. Math. 18 (2015), 1–129.
[47] L. Libkin, Elements of Finite Model Theory. Texts in Theoretical Computer Science. An EATCS Series. Springer-Verlag, Berlin, 2004.
[48] R.C. Lyndon, Identities in finite algebras, Proc. Amer. Math. Soc. 5 (1954), 8–9.
[49] R. McKenzie, Equational bases for lattice theories, Math. Scand. 27 (1970), 24–38.
[50] R. McKenzie: The residual bounds of finite algebras. Internat. J. Alg. Comput. 6 (1996), 1–28.
[51] R. McKenzie: The residual bound of a finite algebra is not computable. Internat. J. Alg. Comput. 6 (1996), 29–48.
[52] R. McKenzie: Tarski's finite basis problem is undecidable. Internat. J. Alg. Comput. 6 (1996), 49–104.

[53] R. Park, Equational classes of non-associative ordered systems, Ph.D. dissertation, UCLA, 1976.
[54] R. Pierce, Homomorphisms of semigroups, Ann. Math. 59 (1954), 287–291.
[55] O. Reingold, Undirected st-connectivity in log-space. In Harold N. Gabow and Ronald Fagin, editors, STOC, pages 376–385. ACM, 2005.
[56] O. Reingold, Undirected connectivity in log-space, J. ACM 55(4) (2008), 1–24.
[57] B. Rossman, Homomorphism preservation theorems, J. ACM 55 (2008), no. 3, Art. 15, 53 pp.
[58] E. Rosen, Some aspects of finite model theory and finite structures, Bull. Symb. Logic 8 (2002), 380–403.
[59] M.V. Sapir, Sur la propriété de base finie pour les pseudovariétés de semigroupes finis. (French) [On the finite basis property for pseudovarieties of finite semigroups] C. R. Acad. Sci. Paris Sér. I Math. 306 (1988), 795–797.
[60] S. Shaheen and R. Willard, Algebras from finite group actions and a question of Eilenberg and Schützenberger, J. Algebra (2023), doi: https://doi.org/10.1016/j.jalgebra.2023.05.014
[61] B.M. Schein, Homomorphisms and subdirect decompositions of semigroups, Pacific J. Math. 17 (1966), 529–547.
[62] Z. Székely, Computational complexity of the finite algebra membership problem for varieties, Internat. J. Algebra Comput. 12 (2002), 811–823.
[63] W.W. Tait, A counterexample to a conjecture of Scott and Suppes, J. Symbolic Logic 24 (1959), 15–16.
[64] V.I. Tumanov, On finite lattices having no independent bases of quasi-Identities, Math. Notes 36 (1984), 625–634.
[65] M.V. Volkov, S.V. Goldberg, and S.I. Kublanovsky, J. Math. Sci. 177 (2011), 847–859.
[66] R. Willard, On McKenzie's method, Periodica Mathematica Hungarica 32 (1996), 149–165.
[67] R. Willard, Tarski's finite basis problem via $\mathbf{A}(\mathcal{T})$, Trans. Amer. Math. Soc. 349 (1997), 2755–2774.
[68] R. Willard, A finite basis theorem for residually finite, congruence meet-semidistributive varieties, J. Symbolic Logic 65 (2000), 187–200.
[69] R. Willard, The finite basis problem, Contributions to General Algebra, vol. 15, 199–206, Heyn, Klagenfurt, 2004.